

Quest Align: Biomechanical Analysis and Real-Time Postural Monitoring

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Abstract—This paper presents Quest Align, an engineering prototype for biomechanical posture monitoring during immersive virtual-reality interaction. The system uses a consumer head-mounted display, hand controllers, and an optional smartphone inertial measurement unit placed near the pelvis. A pretrained HMD-Poser model is used as a sparse-motion reconstruction backbone, while a custom runtime layer converts incoming sensor packets into model-compatible features and reduces the predicted SMPL trunk chain into interpretable pitch and lateral-roll indicators. The implementation receives JSON packets over UDP, synchronizes headset and smartphone streams by server-side timestamps, normalizes quaternions, constructs a fixed-size feature tensor, and computes posture feedback after neutral calibration and median filtering. The prototype stores per-session JSON records that include sensor metadata, inferred angles, calibration offsets, and alert flags. The current work should be interpreted as a functional software and systems-integration prototype rather than a clinical posture-assessment device, because no synchronized optical motion-capture, inclinometer, or radiographic ground truth was collected. Nevertheless, the results show a feasible path for using VR hardware as a low-intrusion platform for real-time postural awareness.

Keywords—virtual reality, posture monitoring, HMD-Poser, SMPL, inertial measurement unit, biomechanics, sparse motion tracking.

I. INTRODUCTION

Postural monitoring is relevant in ergonomics, rehabilitation, human-computer interaction, sports training, and extended computer use. Poor trunk alignment during repetitive tasks may be difficult to notice by the user, especially when attention is focused on a screen or on an immersive task. Conventional posture assessment can involve observation by a specialist, manual inclinometers, marker-based motion capture, depth cameras, or body-worn inertial measurement units (IMUs). These tools can be useful, but many of them require external equipment, fixed acquisition spaces, or a preparation time that is not ideal for ordinary virtual-reality (VR) sessions.

Modern VR systems already include sensors that estimate the position and orientation of the head and hands. These measurements are designed mainly for interaction and avatar animation, but they also provide a compact description of the upper body. The main difficulty is that the spine, pelvis, and lower limbs are not directly observed in a standard headset-and-controller setup. Estimating trunk posture from sparse data is therefore an under-constrained problem: the

same head and hand configuration may correspond to several possible spine and pelvis configurations. Learning-based full-body reconstruction methods address this limitation by using motion priors learned from large motion-capture datasets [1]–[3].

Quest Align explores whether this type of sparse reconstruction can be redirected toward real-time postural awareness. The project does not attempt to replace clinical instruments. Instead, it implements a complete acquisition and inference pipeline that transforms consumer VR signals into coarse biomechanical indicators. The name Quest Align emphasizes two ideas: the use of a Meta Quest-style sensing environment and the goal of estimating alignment-related variables in real time.

The specific contributions of this paper are fourfold. First, it describes a two-stream runtime architecture that receives headset/controller data and optional smartphone IMU data through UDP. Second, it documents the feature-construction step that adapts the available signals to a pretrained HMD-Poser network. Third, it proposes a simple trunk-angle extraction module based on the SMPL spine chain. Fourth, it defines a session logging structure that preserves raw measurements, calibration values, inferred angles, and alert states for later analysis.

II. BACKGROUND AND RELATED WORK

A. Sparse VR Motion Estimation

Sparse body tracking estimates unobserved joints from a small number of measured body parts. In mobile VR, the usual input is the six-degree-of-freedom motion of the head and hands. AvatarPoser showed that articulated full-body pose can be inferred from sparse motion sensing, making full-body avatar animation possible with fewer trackers [2]. HMD-NeMo addressed online avatar motion generation from sparse head-mounted-display observations and emphasized latency-aware inference for interactive scenarios [3]. SparsePoser investigated real-time reconstruction from a reduced tracker set and demonstrated the value of learned motion priors in sparse tracking tasks [4].

HMD-Poser is the closest technical basis for the present system. It was designed to recover full-body motion from scalable sparse observations, including HMD-only and HMD-plus-

IMU configurations [1]. Its architecture combines temporal and spatial feature learning to operate in real-time settings. Quest Align reuses this model as an inference backbone, but changes the application objective: the final output is not only an avatar pose, but also a reduced set of trunk posture variables.

B. Wearable Posture Monitoring

Wearable posture systems have been investigated for rehabilitation support, exercise feedback, and everyday musculoskeletal monitoring. Simpson *et al.* reviewed wearable technologies for spinal posture analysis and reported that live feedback and inertial sensing are common directions in this field [5]. Earlier trunk-monitoring studies used inertial sensors to estimate orientation changes in the sagittal and frontal planes [6]. Other systems implement real-time bad-posture alerts or sensor-array feedback during exercise [7], [8].

Compared with these approaches, Quest Align reduces the number of dedicated body-worn sensors. The headset supplies the most stable spatial reference, the controllers provide upper-body context, and the smartphone IMU acts as an optional pelvis cue. This design is less anatomically detailed than a multi-IMU spine system, but it may be more convenient for applications where the user already wears a VR headset.

C. Body Model and Rotation Representation

The body representation used by Quest Align follows the SMPL kinematic structure because HMD-Poser predicts SMPL-compatible joint motion. SMPL is a skinned parametric body model that represents human shape and pose in a graphics-compatible formulation [9]. AMASS provides a large archive of motion-capture sequences expressed as surface shapes and SMPL parameters [10], which makes it useful for training and evaluating full-body reconstruction methods. The pose output uses 6D rotation representations, which are preferred in several neural settings because they avoid discontinuities that can occur with some lower-dimensional rotation encodings [11].

III. SYSTEM DESIGN

A. Design Requirements

The prototype was designed around five practical requirements. The first requirement is low setup complexity: a headset, its controllers, and a smartphone should be sufficient to run a session. The second requirement is real-time operation, because posture feedback is only useful if the user receives it during the activity. The third requirement is robustness to partial sensing; the system should continue running even when the phone stream is unavailable. The fourth requirement is reproducible logging, so that each session can be inspected after acquisition. The fifth requirement is explicit separation between functional monitoring and clinical diagnosis.

These requirements guided the architecture shown in Fig. 1. The pipeline has four functional stages: acquisition, synchronization, feature construction, and posture reduction. Each stage is intentionally simple so that failure points can be identified during laboratory testing.

TABLE I
MAIN SOFTWARE MODULES IN THE PROTOTYPE

Module	Role in Quest Align
<code>server.py</code>	Coordinates sensor streams, synchronization states, calibration, inference, alerts, and JSON session writing.
<code>udp_listener.py</code>	Opens UDP sockets and forwards received packets to stream-specific handlers.
<code>feature_builder.py</code>	Normalizes quaternions, converts them to rotation matrices, assembles sensor vectors, and applies zero padding.
<code>spine_analyzer.py</code>	Computes posture-related angle estimates from SMPL trunk joint coordinates or rotations.
Notebook	Tests loading, simulated inference, visualization, and angle-extraction behavior.

B. Hardware and Software Roles

The headset-side stream provides the position and orientation of the head, the left hand, and the right hand. The Android-side stream provides a pelvis quaternion and a linear acceleration vector. In the implemented code, the headset stream uses UDP port 5006 and the phone stream uses UDP port 5005. The server runs a main loop configured at 60 Hz and tracks the most recent timestamp for each stream.

The software modules are mapped in Table I. The server controls the acquisition state machine, packet buffers, calibration logic, and session storage. The UDP listener isolates network input from the main loop. The feature builder performs the mathematical conversion from raw sensor values into the model input. The spine analyzer and server-side angle functions then reduce model output to posture variables.

IV. MATERIALS AND METHODS

A. Acquisition and Synchronization

Each incoming packet is decoded as JSON. The headset packet contains position and quaternion entries for the head and both hands. The Android packet may be a control message, a clock-synchronization request, or an IMU sample. Only IMU samples are inserted into the synchronized data buffer. To avoid mixing incompatible clock origins, the current server uses packet arrival time as the synchronization reference for both streams.

A sensor is treated as inactive when no packet has arrived for 0.5 s. If both streams are active, the system enters a 3 s warm-up stage and then records synchronized pairs whose arrival timestamps differ by no more than 100 ms. If the phone stream is not active, the program can continue in Quest-only mode by inserting a neutral pelvis quaternion and a gravity-like acceleration prior. This fallback prevents software interruption but should not be interpreted as a reliable pelvis measurement.

B. Feature Construction

All quaternions are normalized and converted to the $[x, y, z, w]$ convention used by the rotation utilities. For each

Real-time spinal posture assessment pipeline

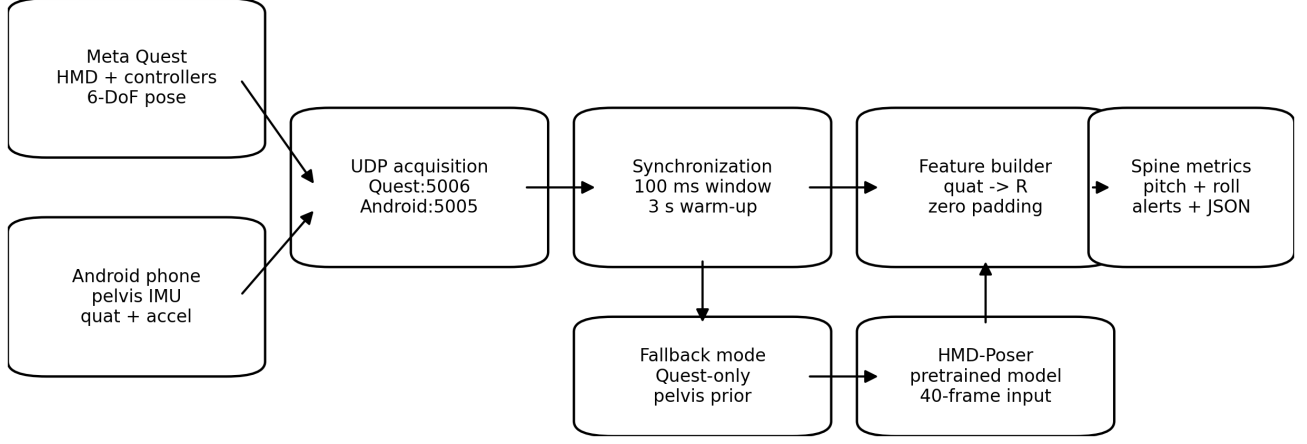


Fig. 1. Runtime pipeline used by Quest Align. The system combines headset/controller data and optional smartphone IMU data, synchronizes packet streams, constructs HMD-Poser-compatible features, and converts predicted trunk motion into pitch/roll feedback.

TABLE II
RUNTIME PARAMETERS USED IN THE PROTOTYPE

Parameter	Value
Headset UDP port	5006
Android UDP port	5005
Main loop frequency	60 Hz
Sensor timeout	0.5 s
Synchronization tolerance	100 ms
Warm-up period	3 s
Calibration window	30 frames
Median filter window	7 frames
Alert threshold	15° for pitch or roll
Minimum saved session	10 valid frames

active sensor k , the feature vector concatenates position, flattened rotation matrix, and acceleration:

$$\mathbf{s}_k = [\mathbf{p}_k, \text{vec}(\mathbf{R}_k), \mathbf{a}_k] \in \mathbb{R}^{15}. \quad (1)$$

Head and hand accelerations are set to zero in this implementation because the available headset stream only provides pose. The pelvis vector uses smartphone orientation and acceleration. The absent lower-leg sensors are represented by zeros. This zero-padding strategy preserves the input dimensionality expected by the pretrained model while making the missing observations explicit.

The full vector is padded or truncated to 135 scalar values and reshaped as a single-frame model input. Because the pretrained model expects a temporal sequence, the current prototype repeats the most recent frame across a 40-frame window. This is a practical engineering adaptation for live testing. A future version should replace this repetition with a rolling buffer containing the last 40 synchronized frames, which would preserve short-term motion dynamics instead of approximating them.

C. HMD-Poser Inference

The pretrained HMD-Poser checkpoint is loaded in evaluation mode. At each inference step, the model receives the constructed temporal tensor and predicts local SMPL pose parameters. The server selects the last predicted frame and interprets it as a 22×6 matrix, where each row corresponds to one SMPL joint represented by a 6D local rotation. Only the trunk-related rows are used in the posture analysis, because the aim is not to reconstruct every limb measurement but to extract a compact summary of spinal alignment.

D. Trunk Angle Extraction

The posture module focuses on the chain Pelvis, Spine1, Spine2, Spine3, Neck, and Head. In the implemented runtime path, a simplified trunk orientation is obtained from the ordered product of the three local spine rotations:

$$\mathbf{R}_T = \mathbf{R}_{\text{Spine1}} \mathbf{R}_{\text{Spine2}} \mathbf{R}_{\text{Spine3}}. \quad (2)$$

The combined rotation is mapped to Euler angles using the same convention as the server implementation. The pitch component is interpreted as a sagittal bending indicator and the roll component as a lateral-deviation indicator. Fig. 2 illustrates the selected trunk chain. The method intentionally avoids claiming vertebra-level measurements, because SMPL spine joints are coarse animation joints rather than anatomical vertebral landmarks.

E. Calibration, Filtering, and Alerts

The first 30 valid inference frames are used to estimate neutral pitch and roll offsets. After calibration, each new value is offset-corrected and passed through a seven-frame median filter. The median filter was chosen because it suppresses isolated frame-level spikes without requiring a complex motion model. The alert logic is intentionally transparent: pitch above 15° indicates forward flexion, pitch below -15° indicates

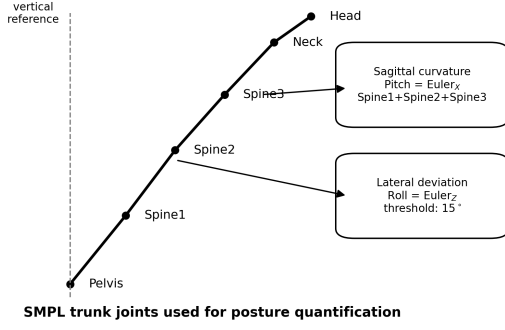


Fig. 2. SMPL trunk chain selected for posture quantification. Quest Align summarizes the trunk with pitch and lateral roll instead of estimating vertebra-level clinical variables.

backward extension, and absolute roll above 15° indicates lateral deviation.

The real-time feedback should be read as a posture-awareness cue. For example, a repeated lateral-roll warning can indicate that the user is leaning to one side during a VR activity. It does not diagnose scoliosis, kyphosis, lordosis, or any musculoskeletal disorder. Clinical interpretation would require validated anatomical landmarks, calibration procedures, and reference measurements.

V. PROTOTYPE EVALUATION

A. Functional Verification

The evaluation performed in this stage verifies whether the pipeline works end to end. The tests covered packet parsing, sensor-state transitions, synchronization-window behavior, quaternion normalization, feature-vector construction, loading of the pretrained model, inference on the constructed tensor, angle extraction, calibration, median filtering, alert assignment, and JSON output generation. These checks confirm that the software can receive live signals, transform them into the model format, and produce posture feedback without manual intervention during a recording session.

The notebook was used to test isolated mathematical components before integration. In particular, simulated trunk configurations were used to verify that neutral and tilted postures produce consistent changes in the expected angular channels. This step does not establish clinical accuracy, but it reduces the risk that the runtime alerts are caused by sign errors, quaternion-format mismatches, or incorrect joint selection.

B. Outputs and Data Products

Each session produces structured outputs suitable for later inspection. Table III summarizes these data products. The most important output is the per-frame inference log, because it contains the time series of pitch and lateral roll. The summary fields are useful for quick reports, while raw sensor values allow debugging when unexpected alerts occur.

TABLE III
OUTPUTS GENERATED BY QUEST ALIGN

Output	Content
Sensor log	Headset, hand, and pelvis measurements with timestamps.
Inference log	Pitch and roll estimates for valid frames.
Session summary	Mean, minimum, and maximum posture values.
Calibration data	Neutral offsets computed from early frames.
Alert flags	Boolean states for forward, backward, and lateral deviations.
System metadata	Frame count, mode, synchronization status, and estimated rate.

C. Expected Feasibility and Boundary of Claims

The underlying real-time feasibility is supported by prior HMD-Poser experiments, but the present prototype does not inherit all validation results from that paper. HMD-Poser validates full-body motion reconstruction under its own experimental protocol [1]; Quest Align adds a new downstream interpretation layer for posture feedback. That layer must be validated separately because errors in trunk angle estimation can arise from sensor placement, model adaptation, calibration, and anatomical simplification.

Therefore, the correct interpretation of the current result is functional feasibility. The system can acquire, infer, filter, warn, and save measurements. It cannot yet report accuracy, sensitivity, specificity, or clinical agreement. A quantitative study would require synchronized reference data and a protocol with controlled postures, repeated participants, and comparisons against accepted measurement instruments.

VI. DISCUSSION

Quest Align demonstrates that a VR headset can be used as the central sensor in a posture-monitoring pipeline. The approach is attractive because it does not require external cameras and can operate inside the same interaction loop used by a VR application. The smartphone pelvis unit adds context without requiring specialized wearable hardware. This makes the prototype appropriate for educational demonstrations, early ergonomic feedback studies, and laboratory exploration of HMD-based biomechanics.

The most important limitation is anatomical resolution. SMPL was not designed as a clinical spine model, and the trunk chain contains only a few joints. As a result, Quest Align estimates coarse trunk orientation rather than vertebral alignment. The second limitation is temporal adaptation. Repeating a single frame across the model window allows live inference to run, but it weakens the temporal information that HMD-Poser expects. A rolling buffer should improve consistency during dynamic movement. The third limitation is sensor placement. A smartphone near the pelvis is convenient, but its orientation depends on how it is attached to the body. Poor fixation can introduce rotations unrelated to the user's pelvis.

TABLE IV
PROPOSED VALIDATION PROTOCOL FOR FUTURE EXPERIMENTS

Validation aspect	Suggested measurement
Synchronization	Time offset between headset and phone samples during repeated movements.
Angular agreement	Mean absolute error against optical motion capture, validated IMUs, or calibrated inclinometers.
Dynamic stability	Jitter and dropped-frame rate during walking, leaning, and seated tasks.
Sensor placement	Error change when the phone is placed at different pelvis or waist locations.
User feedback	Comfort, distraction, and perceived usefulness of alerts during VR tasks.

Another limitation is threshold generality. The selected 15° alert threshold is useful for prototyping, but it should not be treated as a universal clinical threshold. Different tasks, users, and body proportions may require different thresholds. A future calibration stage could ask the user to perform neutral, forward-flexed, backward-extended, and lateral-leaning poses, then adapt thresholds to that individual baseline.

VII. FUTURE WORK

The next version should replace the repeated-frame temporal input with a rolling sequence buffer and should align the feature builder more closely with the official HMD-Poser preprocessing pipeline. A second priority is collecting a validation dataset. Each participant should perform predefined postures while the VR system records headset/controller data, the smartphone records pelvis IMU data, and a reference system records ground truth. Possible references include optical motion capture, validated multi-IMU systems, or calibrated inclinometers. The comparison should report angular error, latency, dropped-frame rate, and robustness to phone placement.

User-facing feedback also requires study. Alerts that are too frequent may distract users inside VR, while alerts that are too rare may fail to correct posture. Future experiments should compare continuous numeric feedback, simple warning icons, auditory cues, and end-of-session summaries. Privacy should also be considered because motion traces can contain identifiable behavioral patterns. Local processing and minimal storage are preferable when the goal is personal posture feedback.

Table IV summarizes a possible validation plan for converting the current prototype into a quantitatively evaluated system. The plan separates technical tests, biomechanical agreement tests, and user-facing tests because each one answers a different question. A system can be fast and stable but still anatomically inaccurate; likewise, a system can be accurate in a laboratory but distracting or uncomfortable during an immersive task.

VIII. CONCLUSION

This paper presented Quest Align, a real-time prototype that links VR headset tracking, optional Android pelvis sensing,

HMD-Poser inference, and SMPL-based trunk-angle extraction. The system receives UDP sensor streams, constructs model-compatible features, estimates pitch and lateral roll, applies calibration and median filtering, and stores structured session records. The work supports the feasibility of HMD-centered posture monitoring as an engineering prototype. Its main value is not clinical diagnosis, but a reproducible implementation path for turning sparse VR tracking into real-time postural awareness. Future validation must compare the inferred indicators against ground-truth measurements before the system can be used in health-related decision-making.

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